

Development of a BI-Rate Prediction System Based on Explainable Artificial Intelligence (XAI) and a Large Language Model (LLM) Using Macroeconomic Indicators

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Executive summary. Six models predict Indonesia's policy rate from 12 macroeconomic indicators, SHAP and LIME attribute every prediction, and GPT-4o narrates that attribution under strict grounding. Aggregate accuracy turns out to be inherited from interest-rate smoothing; the usable signal sits in the nine months when the rate actually moves.

251
monthly rows

384
combinations

12
indicators

6
models

17
features

3,99
expert score

PART 1 Problem, Contribution, Data

A public decision with no public account of its drivers

01 Why It Is a Black Box

- 1

Opaque decision
The rate moves in 0.25–point steps and anchors inflation, the exchange rate, and credit. No quantitative account of each indicator's contribution exists.
- 2

Opaque models
Machine learning captures the nonlinear interaction econometrics misses, but cannot itself be audited.
- 3

Unreadable attribution
SHAP and LIME return plots and matrices a non-technical stakeholder cannot act on.

In public policy, accuracy alone is insufficient. The explanation has to be checkable line by line.

02 Contributions

- 1

Factorial benchmark. 384 preprocessing combinations across six models in a locked results registry.
- 2

Persistence-aware protocol. Change and hold months scored apart, corroborated by leave-one-out ablation.
- 3

Two-layer transparency stack. Exact TreeSHAP with LIME as a direction check, narrated by a grounded LLM.
- 4

Expert validation. A Bank Indonesia monetary economist scoring a 96–item instrument.

RESEARCH QUESTIONS

- 1

How can accurate predictions be produced from nonlinear domestic and global indicators?
- 2

How can a black box prediction be explained to non-technical stakeholders?

03 Data and Preprocessing

251 monthly rows, July 2005 to May 2026. 12 indicators from Bank Indonesia, Statistics Indonesia, FRED, CBOE, and Yahoo Finance: inflation and its gap, GDP growth, M2, credit, USD/IDR, equity index, reserves, Federal Funds Rate, Brent, gold, VIX.

MARGINAL RMSE RANGE PER FACTOR

- 1

Feature engineering, policy-reaction block

0.7444
- 2

Feature selection, expert-curated set

0.6892
- 3

Imputation, causal forward fill wins

0.4275
- 4

Scaling, negligible for tree models

0.0009

The three leading models share one recipe: forward fill, expert-curated features, policy-reaction engineering.

PART 2 The System

Six stages, each re-runnable from the locked registry

STAGE 1

Data
251 rows, 12 indicators, five official sources merged.

STAGE 2

Preprocessing
Forward fill, chronological 60/20/20, scalars fit on train only.

STAGE 3

Features
17 policy-reaction features: lags, real rate, output gap, stance.

STAGE 4

Training
ARIMA, VAR, RF, XGBoost, LSTM, BiLSTM, Optuna TPE, three-seed ensembles.

STAGE 5

Attribution
Exact TreeSHAP, KernelExplainer for networks, LIME as direction check.

STAGE 6

Narration
GPT-4o writes the report, an expert scores it on 96 items.

04 Grounding the Narrative

FOUR RULES ON THE GPT-4O LAYER

- 1

The payload carries only SHAP values and that month's observed indicator movements.
- 2

No fact outside the payload may enter the report.
- 3

Magnitudes as a share of total absolute contribution, directions as pushing up or holding down, never as causation.
- 4

Every number in the narrative resolves to a row of the SHAP table.

July 2025, Random Forest: the one-month lag carries **73%** of the month's influence and credit growth **15%**, both traceable to local SHAP values.

05 Predictive Performance

MODEL	RMSE	MAPE	R ²
Random Forest	0.1434	1.95	0.8911
BiLSTM	0.1463	2.01	0.8867
LSTM	0.1620	2.30	0.8610
XGBoost	0.2207	3.18	0.7420
ARIMA	0.8974	14.05	−3.265
VAR	2.7089	38.90	−37.868

- 1

Machine learning beats econometrics by an order of magnitude. That is the only reproducible hierarchy.
- 2

The top three sit 0.003 to 0.019 apart against inter-run deviations of ±0.009 and ±0.015, so no single winner is claimed.

06 The Ranking Reverses

27 hold months and 9 change months, scored apart

MODEL	CHANGE	HOLD	DIR.
Random Forest	0.214	0.110	33%
BiLSTM	0.162	0.141	67%
LSTM	0.179	0.156	67%
XGBoost	0.241	0.213	67%

- 1

Random Forest earns its headline position on hold months, where copying last month's value is nearly optimal.
- 2

The recurrent models take persistence as an anchor outside the feature matrix, so they learn the macroeconomic correction instead.
- 3

Nine change months move a hit rate about 11 points each, so ablation has to corroborate the reversal.

PART 3 Where Accuracy Comes From

Ablation, attribution, expert judgement

07 Ablation

Test RMSE per variant

VARIANT	RF	BiLSTM	LSTM	XGB
Full configuration	0.1434	0.1463	0.1620	0.2207
Without BI-Rate t-1	0.4087	0.1378	0.1689	0.2563
Without macro block	0.1272	0.1508	0.1989	0.2030
Lag only, three features	0.1174	0.1172	0.1205	0.1631
Without residual anchoring	—	0.4876	0.3331	—

- 1

Removing one lag nearly triples Random Forest error while both recurrent models stay inside the noise band.
- 2

The lag-only variant wins on aggregate metrics, yet its change-month direction collapses to 0 to 22 percent. It is a smoothed copy of last month's rate.
- 3

Residual anchoring is the decisive design component for the recurrent models.

08 Attribution

Random Forest, July 2025 control month



- 1

Persistence leads with mean [SHAP] 0.9611, a quantitative confirmation of interest-rate smoothing.
- 2

Beyond it, credit growth 0.0307, the inflation gap 0.0097, and the policy stance 0.0089 lead an economically coherent ordering.
- 3

Attribution does not transfer: persistence is 75.5% of Random Forest but 6.7% of LSTM.
- 4

LIME agrees only on persistence, median surrogate R² 0.0247, so it stays a direction check.

09 Expert Validation

96 items, Bank Indonesia monetary economist

OVERALL 3.99 of 5, zero hallucination findings	4.21 LSTM	4.00 BiLSTM
	3.92 Random Forest	3.83 XGBoost

- 1

Absence of hallucination and narrative-to-SHAP consistency scored 5 of 5 throughout; lay readability scored 3.
- 2

The expert ranking does not follow the accuracy ranking, because the most accurate model's attribution collapses onto persistence.
- 3

Both extremes of the feature-importance item are reproducible from ablation obtained independently of the questionnaire.

CONCLUSION

Aggregate accuracy is inherited from interest-rate smoothing, which the system itself quantifies. The predictive value is real but narrow, concentrated in the months when the rate moves. For policy analysis the system is usefulness as a forecaster than as an audit surface: every claim in the narrative resolves to a SHAP value the reader can check line by line.

LIMITATIONS AND NEXT STEPS

- 1

A single rater, so no inter-rater statistics are possible. Multi-expert validation is the next step.
- 2

Only nine change months exist in the test window.
- 3

The narrative layer calls a proprietary API; a central-bank deployment should host it locally.